

PROBLEM STATEMENT & OBJECTIVE DEFINITION

Team ID: SWTID-2026-3238 | Project Name: PocketSmart AI | Track: AI-ML & Generative AI

1. Background & Problem Context

Consumers frequently face budgeting friction when planning complex lifestyle purchases such as home interiors, parties, and personal fashion. Current digital shopping tools optimize for maximizing order value rather than helping users stay strictly within their defined budget limits.

2. Formal Problem Statements

- **PS-1 (Manual Overhead):** Users spend excessive hours manually calculating budget allocations across furniture, catering, decor, and accessories without real-time recommendations.
- **PS-2 (Multimodal Gap):** Users struggle to visually match accessories (jewelry/styling) to specific outfits under strict financial constraints.
- **PS-3 (Budget Enforcement):** Existing AI tools produce hallucinations or generic advice that disregard exact cost constraints.

3. Core Project Objectives

- Build a lightweight, high-performance web app using FastAPI, Jinja2, and SQLite.
- Integrate Google Gemini 2.5 Flash for multimodal vision and natural language recommendations.
- Ensure zero-downtime offline continuity via a deterministic mathematical fallback engine.